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# Decision Tree for Spam Detection:
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.feature_extraction.text import CountVectorizer
df=pd.read_csv("email.csv")
x=df["Message"]
y=df["Category"]
le=LabelEncoder()
y=le.fit_transform(y)
cv = CountVectorizer()
x = cv.fit_transform(x)
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
model=DecisionTreeClassifier(criterion="gini",max_depth=3)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
print(y_pred)

# Random Forest for Spam Detection:
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_extraction.text import CountVectorizer
df=pd.read_csv("email.csv")

x=df["Message"]
y=df["Category"]

le=LabelEncoder()
y=le.fit_transform(y)

cv = CountVectorizer()
x = cv.fit_transform(x)
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x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)

model=RandomForestClassifier(criterion="gini",n_estimators=10)
model.fit(x_train,y_train)

y_pred=model.predict(x_test)
print(y_pred)
```

Outputs:

```
(.venv) PS D:\Internship\Day41> python task.py
[0 0 0 ... 1 0 0]
[0 1 0 ... 0 0 0]
```